1.(i) Write a R program to create three vectors a,b,c with 3 integers. Combine the three vectors to become a 3×3 matrix where each column represents a vector. Print the content of the matrix.

(ii) Write a R program to create three vectors numeric data, character data and logical data. Display the content of the vectors and their type.

2.What are the different forms of data types and how to test the data type in R? Give one example for each.

3. Create a dm matrix which is 3×3 matrix and each row represents a following vector a = c(1, 2, 3), b = c(4, 5, 6), c=c(7, 8, 9) and find the value of the following

a.) How do I select the c (4, 5, 6)?

b.) How do I select the 8?

c.) What is dm[2, 2]?

d.) What is dm [, 3]?

e.) What is dm[1,]?

4. Write in Detail about various Advanced-Data structures with example. .

5.(i) Create vector of numeric, complex, logical and character with types of length 6.

(ii) Write a R program to add a new item g4 = "R Prog" to a given list.
given list: (g1 = 1:20, g2 = "Python", g3 = "HTML").

6. The price of one kg of rice is Rs. 40.75 and one kg of sugar is Rs. 30. Write R program to get the total amount of 2kg rice and 5kg sugar purchase.

7. Create an Array with name “MySales” with 30 observations using following methods: a) By using the array with dimensions 3, 5 and2. b) By using Vector method.

8. (a) Write a R code to create a 3x4 matrix with 12 random numbers between 1-100; have the matrix be filled our row-by-row, instead of column-by-column. Name the columns of the matrix uno, dos, tres, cuatro, and the rows x, y, z. Scale the matrix by 10 and save the result. (7M)

(b) Extract the column called “uno” as a vector from the original matrix and save the result (3M)

9. (a)Write a R program to add a new item g4 = "R Prog" ,g5= “C++” to a given list.

Sample list: (g1 = 1:20, g2 = "Python", g3 = "HTML")

(b). Write a R program to extract all elements except the second element of the first vector of a given list.

Sample list: (g1 = 1:20, g2 = " Python ", g3 = "HTML",g4="R Prog", g5=”C++”)

10. Consider A=matrix(c(2,0,1,3),ncol=2) and B=matrix(c(5,2,4,-1),ncol=2).

(i) Write a R code a)Find A + B b) Find A – B

(ii) Find the Matrix multiplication of A*B, and write the R Code

11. Create a user defined function fact(j) to return the the factorial of j using functions in R.

12.a. Write R code to return a complex object

b. How to pass default values for arguments in R

13. Write R function that computes Sum, average, median of given vector

14. Explain the functioning of apply(), lapply() and sapply() in R program with one example each.

15. Write R code function to generate first n terms of a Fibonacci series

16.Write R Program to find maximum and minimum value of a given vector using control statement.

```
nums = c(10, 20, 30, 40, 50, 60)
```

17.(a) Write R function to sort a vector in ascending order using control statements

(b) Write R function that returns minimum value of a vector . function must created using control and loop statements

18. (a)Write a R program to create a sequence of numbers from 20 to 50 and find the mean of numbers from 20 to 60 and sum of numbers from 51 to 91.

(b) Write R code to print the vector elements one by one using “for loop”

19. (a)Write R code to the function by using if else command

$$f(x) = x \text{ if } x<0.5$$
$$=(1-x) \text{ if } 0.5<= x<=1$$
$$=0 \text{ otherwise}$$

(b) Write R function to find sum and mean of the given vector

20. Given 3 linear equations

5x1-x2+3x3=7

3x1+2x2+4x3=10

7x1-4x2+8x3=-14

Solve the linear equations and find x1,x2,x3 values using appropriate R function

21. What is data frame and how to create a data frame using the following data:

Height	GPA
66	3.80
62	3.78
63	3.88
70	3.72
74	3.69

Write a R program to prepare the inventory. find the mean of height and GPA

22. (a)Write a R program to create a data frame using two given vectors and display the duplicated elements and unique rows of the said data frame.

Sample data:

```
a = c(10,20,10,10,40,50,20,30)
b = c(10,30,10,20,0,50,30,30)
```

(b)write R code to find mean of "b" with respect to a in the above data frame

23. (a)Write a R program to save the information of a following data frame in a file and display the information of the file.

Sample data:

```
exam_data = data.frame(
name = c('Anastasia', 'Dima', 'Katherine', 'James', 'Emily', 'Michael', 'Matthew', 'Laura'),
score = c(12.5, 9, 16.5, 12, 9, 20, 14.5, 13.5),
attempts = c(1, 3, 2, 3, 2, 3, 1, 1),
qualify = c('yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no'))
```

(b)Write R code which display the average score who clear in first attempt.

24.Write R code to perform the operations with following data

```
exam_data = data.frame(
name = c('Anastasia', 'Dima', 'Katherine', 'James', 'Emily', 'Michael', 'Matthew', 'Laura'),
score = c(12.5, 9, 16.5, 12, 9, 20, 14.5, 13.5),
attempts = c(1, 3, 2, 3, 2, 3, 1, 1),
qualify = c('yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no'))
```

(a) Create a Data frame (2M)
(b) Extract score and attempts column by their names (2M)
(c) Extract name and qualify column by their postions (2M)
(d) To extract 3rd and 5th rows with 1st and 3rd columns (2M)
(e) To extract 'James' details (2M)

25. Explain melting and casting data in R with examples

26. Create Data Frame for following data

n <- c(1, 1, 2, 2), time <- c(1, 2, 1, 2) , x <- c(6, 3, 2, 5), y <- c(1, 4, 6, 9)

(i)Write a R code to melt the data and display as a long-format data?

(ii) Write a R code, use cast function appropriately to compute the average of x and y with respect to "time".

27. (a)Write a R program to sort a given data frame by name and score columns
(b)Write R program to find a average score based on "qualify"

```
exam_data = data.frame(
name = c('Anastasia', 'Dima', 'Katherine', 'James', 'Emily', 'Michael', 'Matthew', 'Laura'),
score = c(12.5, 9, 16.5, 12, 9, 20, 14.5, 13.5),
attempts = c(1, 3, 2, 3, 2, 3, 1, 1),
qualify = c('yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no'))
```

28. Write a R program to create inner, outer, left, right join(merge) from given two data frames

```
df1 = data.frame(numid = c(12, 14, 10, 11))
df2 = data.frame(numid = c(13, 15, 11, 12))
```

29. (a)Write a R program to replace NA values with 3 in a given data frame and sort a given data frame by score columns.
(b) Write R program to find a average score who qualified in fisrt attempts

```
exam_data = data.frame(
name = c('Anastasia', 'Dima', 'Katherine', 'James', 'Emily', 'Michael', 'Matthew', 'Laura'),
score = c(12.5, 9, 16.5, 12, 9, 20, 14.5, 13.5),
attempts = c(1, NA, 2, NA, 2, NA, 1, 1),
qualify = c('yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no'))
```

30.a) Write a R program to change column names in a given data frame
(i) "name" to "stu.name". (ii) "score" to "avg.score"

```
name = c('Anastasia', 'Dima', 'Katherine', 'James', 'Emily', 'Michael', 'Matthew', 'Laura'),
score = c(12.5, 9, 16.5, 12, 9, 20, 14.5, 13.5),
attempts = c(1, NA, 2, NA, 2, NA, 1, 1),
qualify = c('yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no'))
```

b) (i)Write a R program to add a new column in the name of “country” and fill the column as per your choice in a given data frame

(ii) Write a R program to add new row(s) to an existing data frame

```
new_exam_data = data.frame(name = c('Robert', 'Sophia'),score =
c(10.5, 9), attempts = c(1, 3),qualify = c('yes',
'no'),country=c(‘USA’, ‘USA’))
```

31. Compute the statistical summary for the following data and write R code for same .Find the Skewness of the data

X<-68 ,64 ,75, 50, 64, 80, 75, 40, 55, 64

32. Compute the statistical summary for the following data and write R code for same Find the Skewness of the data

X<-62, 58, 68, 45, 81, 60, 68 ,48, 58, 70

33.Find the mean ,Variance, 1st Quartile, 2nd Quartile , 3rd Quartile of the given data . Write R code for same

C<-78, 93, 68, 84, 90, 74, 64, 55, 80

34. Explain Covariance and Correlation with example. Infer the relation among attributes with respect to correlation coefficient.

35. Compute the correlation coefficient for the following data and write R code for same

X<-68 ,64 ,75, 50, 64, 80, 75, 40, 55, 64

Y<-62, 58, 68, 45, 81, 60, 68 ,48, 58, 70

36.Explain Skewness and Kurtosis and its types

37. Calculate the coefficient of correlation to the following data

x <-10, 12 ,18, 24, 23, 27

y <-13 ,18, 12 ,25 ,30 ,10

38. Following are the runs scored by a batsman in 10 consecutive matches:

22,98,13,54,77,61,45,32,19,85 Compute coefficient of variation, mean, median, mode. Write R code for same.

39. Suppose a hospital tested the age and body fat data for 18 randomly selected adults with the f ollowing result

age	23	23	27	27	39	41	47	49	50
%fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2

Calculate the standard deviation of age and %fat.

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41. Find the quartiles of this data set: 6, 47, 49, 15, 43, 41, 7, 39, 43, 41, 36. and draw the boxplot . Give the R code for same.

42. Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70. Can you find (roughly) the first quartile (Q1) and the third quartile (Q3) of the data?. Draw the Boxplot with that information. Give the R code for same.

43. a) The maximum temperature in celcius in a week is given as T(35,42,38,25,28,36,40). Draw the bar plot for the given data. Also use legend function to describe hists. (6M)

b) Explain the differences between stacked bar plot and bar plot. (4M)

44. a) Fit a poisson distribution with the following data (6M)

X	0	1	2	3	4	5
F	142	156	69	27	5	1

b) Explain in detail about “lm” functions in R (4M)

45.a) Write R program to plot the function f(x)= sin(x) in the interval (-3,3) in steps of 0.1. The point character of the plot 1's to be triangle joined with the lines.

b) Demonstrate about Box-Whisker Plots

46. a) What is Box plot? Explain importance of boxplot with example?

b) Draw a pie chart for the following data

Section: I, II, III , IV, V

No.of workers:220,370, 190, 70, 250

47. a) Heights(in cm) of father and son are given as follows

Father(X): 150 152 155 157 160 161 164 165

Son (Y) : 154 156 158 159 160 162 161 164

Fit a regression line parameters to predict the height of son given the height of father. Write R code for same.

48. Fit a regression line parameters distribution with the following data

X	0	1	2	3	4	5
Y	142	156	69	27	5	1